

Support Vector Machine

CS21206: Foundations of AI and ML

Abir Das & Ayan Chaudhury

IIT Kharagpur

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Agenda

- § Understanding the decision making in Support Vector Machines (SVM)
- § SVM with soft and hard margins

Resources

- § [Lecture 16](#) of Fall 2010 Artificial Intelligence Course at MIT Prof. Patrick Winston.
- § [SVM Tutorial](#) by Prof. Anoop Namdoodiri, IIIT Hyderabad.

Perceptron Learning

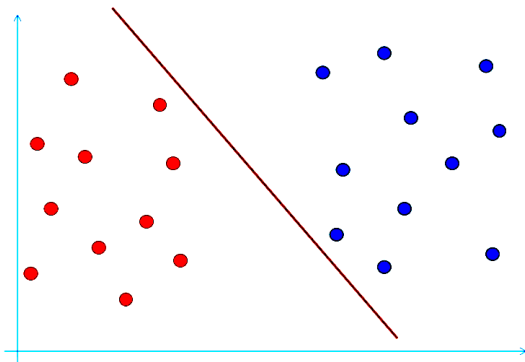


Fig. credit: Prof. Anoop Namboodiri

§ Multiple solutions exist for linearly separable data.

Perceptron Learning

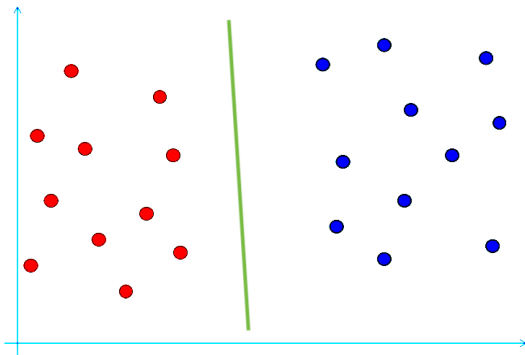


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- § Multiple solutions exist for linearly separable data.
- § Perceptron learning results in **feasible** solution.

Perceptron Learning

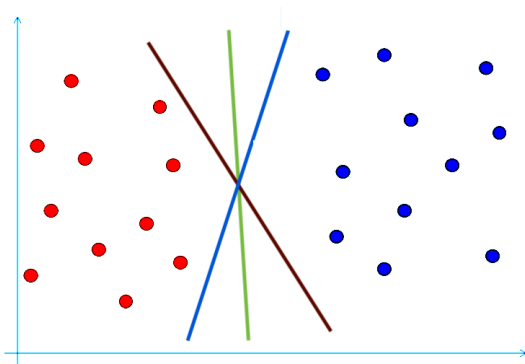
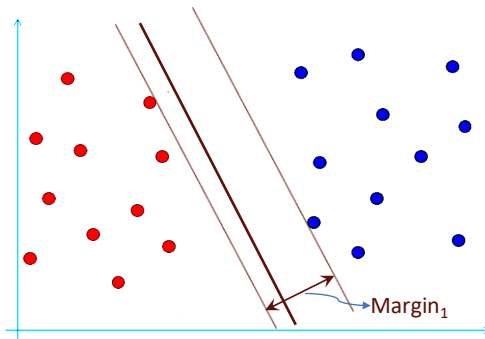


Fig. credit: Prof. Anoop Namboodiri

- § Multiple solutions exist for linearly separable data.
- § Perceptron learning results in **feasible** solution.
- § Are all solutions equally good?

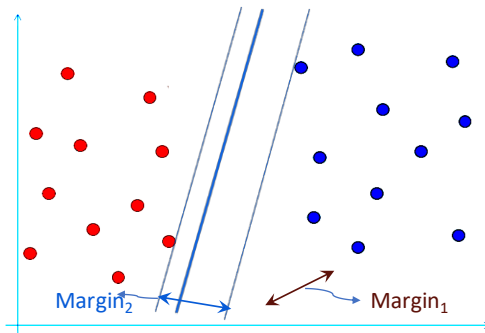
Margin: The No-mans Band



§ Width of a band around decision boundary without any training samples

Slide. credit: Prof. Anoop Namboodiri

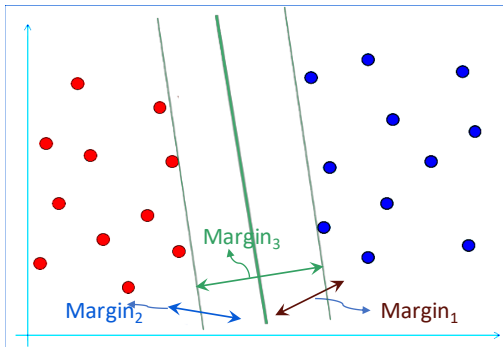
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Margin: The No-mans Band



- § Width of a band around decision boundary without any training samples
- § Margin varies with the position and orientation of the separating hyperplane

Slide. credit: Prof. Anoop Namboodiri

Board Work

- § Here we formulated the optimization problem in the white board.
- § I shall digitize those and add it here, later.
- § Next slide provides the optimization problem directly and we will start discussing its solution strategy thereafter.

The Problem Statement: Hard-Margin SVM

We want to find the hyperplane $(\mathbf{w}, b)$ that maximizes the margin.

The Primal Optimization Problem

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$

subject to:

$$- [y^{(i)} (\mathbf{w}^T \mathbf{x}^{(i)} + b) - 1] \leq 0, \quad \forall i = 1, \dots, N$$

The optimization problem

- § Minimizes a quadratic function.
- § with N linear **inequality constraints**.
- § Simple calculus (setting gradient to zero) doesn't work directly here.

Method of Lagrange Multipliers

- § Converts a *constrained* problem into an *unconstrained* one.
- § For each constraint $g_i(\mathbf{w}, b) \leq 0$, we introduce a new variable $\alpha_i \geq 0$ (the **Lagrange Multiplier**).

The Lagrangian Function $L(\mathbf{w}, b, \alpha)$

$$L(\mathbf{w}, b, \alpha) = \underbrace{\frac{1}{2} \|\mathbf{w}\|^2}_{\text{Objective}} - \sum_{i=1}^N \alpha_i \underbrace{\left[y^{(i)} (\mathbf{w}^T \mathbf{x}^{(i)} + b) - 1 \right]}_{\text{Constraint} \geq 0}$$

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The Min-Max Requirement

To solve the original problem, the Lagrangian formulation demands that:

- 1 We **Minimize** L with respect to the **Primal Variables** $(\mathbf{w}, b)$.
- 2 We **Maximize** L with respect to the **Lagrange Multipliers** $(\alpha_i \geq 0)$.

Mathematically:

$$\min_{\mathbf{w}, b} \left(\max_{\alpha \geq 0} L(\mathbf{w}, b, \alpha) \right)$$

Why Maximize w.r.t. α ? (slight detour)

The Lagrangian Function $L(\mathbf{w}, b, \alpha)$

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Think of α as a **Security Guard** for the constraints:

§ **Case 1: Constraint Violated** i.e., $\mathbf{w}$ is bad: $(y(\mathbf{w}^T \mathbf{x} + b) - 1 < 0)$

- ▶ We don't want such $\mathbf{w}$ to be the answer.
- ▶ $\mathbf{w}$ won't be the answer if L is high.
- ▶ By maximizing $-\alpha \times (\text{Negative Value})$, (by making $\alpha \rightarrow \infty$) we make L very high.

§ **Case 2: Constraint Satisfied** i.e., $\mathbf{w}$ is feasible: $(y(\mathbf{w}^T \mathbf{x} + b) - 1 \geq 0)$

- ▶ Then, we don't want the constraint term to disturb.
- ▶ This is made possible if $-\alpha \times (\text{Positive Value})$ is maximized w.r.t. α as in that case, the maximum of this term is 0 (corresponding to $\alpha = 0$)

Conclusion

Maximizing over α makes any infeasible choice of w, b “expensive.”

The Saddle Point

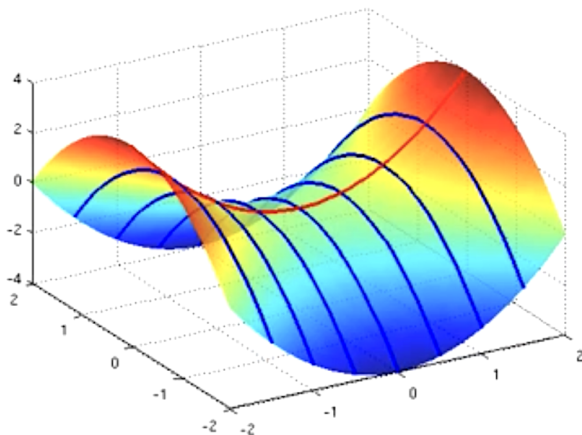


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Karush-Kuhn-Tucker (KKT) Conditions

The “Checklist” at the optimum $(\mathbf{w}^*, b^*)$:

- 1 **Stationarity:** $\nabla_{\mathbf{w}} L|_{\mathbf{w}=\mathbf{w}^*} = 0$ and $\nabla_b L|_{b=b^*} = 0$.
- 2 **Primal Feasibility:** $y_i(\mathbf{w}^{*T} \mathbf{x}^{(i)} + b^*) - 1 \geq 0$.
- 3 **Dual Feasibility:** $\alpha_i \geq 0$.
- 4 **Complementary Slackness:** $\alpha_i [y^{(i)}(\mathbf{w}^{*T} \mathbf{x}^{(i)} + b^*) - 1] = 0$

This implies that α_i can be non-zero, only when $y_i(\mathbf{w}^{*T} \mathbf{x}^{(i)} + b^*) = 1$ i.e., α_i can be non-zero **only** for points on the margin (Support Vectors).

Back to our Formulation

$$L(\mathbf{w}, b, \alpha) = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^N \alpha_i y^{(i)} (\mathbf{w}^T \mathbf{x}^{(i)} + b) + \sum_{i=1}^N \alpha_i$$

At the optimum:

$$1: \frac{\partial L}{\partial \mathbf{w}} = 0; \mathbf{w}^* = \sum_{i=1}^N \alpha_i y^{(i)} \mathbf{x}^{(i)} \quad 2: \frac{\partial L}{\partial b} = 0; \sum_{i=1}^N \alpha_i y^{(i)} = 0$$

KKT Condition (Complementary Slackness):

$$3: \alpha_i [y_i (\mathbf{w}^{*T} \mathbf{x}^{(i)} + b^*) - 1] = 0$$

Regrouping the Lagrangian terms for easier substitution:

$$\begin{aligned} L(\mathbf{w}, b, \alpha) &= \sum_{i=1}^N \alpha_i + \frac{1}{2} \mathbf{w}^T \mathbf{w} - \underbrace{\mathbf{w}^T \sum_{i=1}^N \alpha_i y^{(i)} \mathbf{x}^{(i)}}_{\mathbf{w}^*} - b \underbrace{\sum_{i=1}^N \alpha_i y^{(i)}}_0 \\ &= \sum_{i=1}^N \alpha_i - \frac{1}{2} \mathbf{w}^{*T} \mathbf{w}^* = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y^{(i)} y^{(j)} [\mathbf{x}^{(i)}]^T \mathbf{x}^{(j)} \end{aligned}$$

The Support Vectors

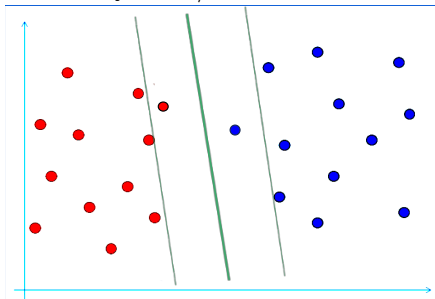
- § Most α_i will be **zero** (points safely away from the boundary).
- § Points with $\alpha_i > 0$ are the **Support Vectors**.

Key Geometric Insight

The entire hyperplane is defined solely by the points on the margin. Moving any other point has **zero effect** on the final model.

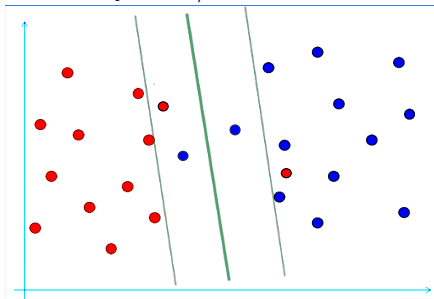
Non-separable Data

Noisy Data/Bad Features



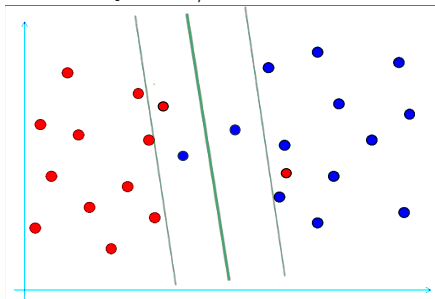
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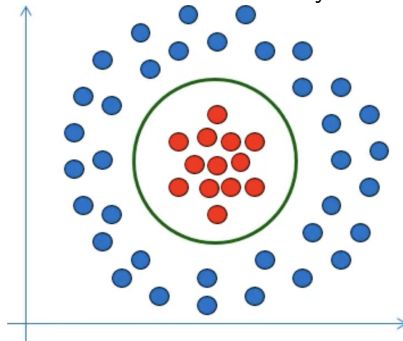


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Non-linear Boundary

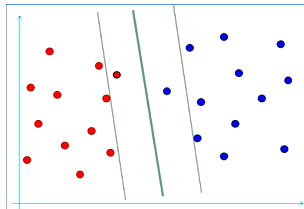


Noise in Data

Optimization Problem (no noise)

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$

$$\text{sub to: } y^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq 1, \forall i$$



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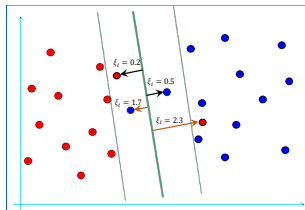
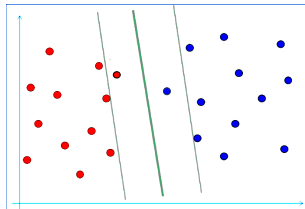
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§ Introduce slack Variables $\xi_i \geq 0$

Optimization Problem (modified)

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$

$$\text{sub to: } y^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq 1 - \xi_i, \forall i$$



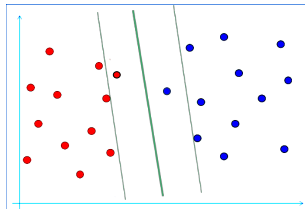
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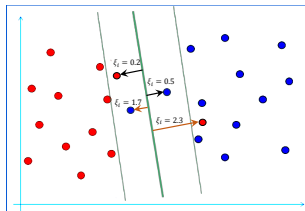
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§ Along with training error also minimize

$$\sum_{i=1}^N \mathbb{1}(\xi_i \geq 1) \text{ or } \sum_{i=1}^N \xi_i$$



Noise in Data

Optimization Problem

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i$$

$$\text{sub to: } y^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq 1 - \xi_i; \xi_i \geq 0, \forall i$$

Forming the Lagrangian and converting to dual, we get,

$$L(\mathbf{w}, b, \boldsymbol{\alpha}) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y^{(i)} y^{(j)} [\mathbf{x}^{(i)}]^T \mathbf{x}^{(j)}$$

$$\text{sub to: } 0 \leq \alpha_i \leq C, \forall i; \sum_{i=1}^N \alpha_i y^{(i)} = 0$$

- § Neither the slack variables ξ_i , nor their multipliers appear in the dual.
- § The only functional change from the hard-margin dual is the additional upper bound C on α_i .